

Algebra 1
Unit 3
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

A line passes through the points $(-8, 1)$ and $(1, 4)$. Use this information for questions 1 through 3.

(x_1, y_1) (x_2, y_2)

1. Find the slope of the line. Show your work.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 1}{1 - (-8)} = \frac{3}{9} = \frac{1}{3}$$

2. Write the equation of the line modeling this table in point-slope form.

Answers may vary. $y - 1 = \frac{1}{3}(x + 8)$

3. Rewrite the equation of the line in standard form.

$$y - 1 = \frac{x}{3} + \frac{8}{3}$$

$$3\left(-\frac{x}{3} + y = \frac{8}{3} + 1\right)$$

$$-x + 3y = 8 + 3$$

$$-x + 3y = 11$$

Patty uses reclaimed rainwater in her garden. At 3 p.m., the water level in her rain barrel was 1.8 inches. During a rain shower, from 3 p.m. to 8 p.m., the rain fell at a steady rate. At 8 p.m., the water level in her rain barrel was 3.9. Use this information for questions 4 through 7.

4. Determine two ordered pairs on the graph of this relationship.

$(0, 1.8)$ and $(5, 3.9)$

5. Use the two ordered pairs to find the slope of the line.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3.9 - 1.8}{5 - 0} = \frac{2.1}{5} = 0.42$$

6. Use the ordered pairs to write the equation of the line in point-slope form.

Answers may vary. $y - 1.8 = 0.42(x - 0)$

7. Use the two ordered pairs to write the equation of the line in slope-intercept form.

$$y - 1.8 = 0.42x$$

$$y = 0.42x + 1.8$$

A line has an x -intercept of 6 and a y -intercept of -2 . Use this information for questions 8 through 10.

8. Find the slope of the line.

$$(6, 0) \quad (0, -2)$$

$$\frac{-2 - 0}{0 - 6} = \frac{-2}{-6} = \frac{1}{3}$$

9. Write the equation of the line in slope-intercept form.

$$y = \frac{1}{3}x - 2$$

10. Write the equation of the line in standard form.

$$\left(-\frac{1}{3}x + y = -2\right) 3$$

$$-x + 3y = -6$$